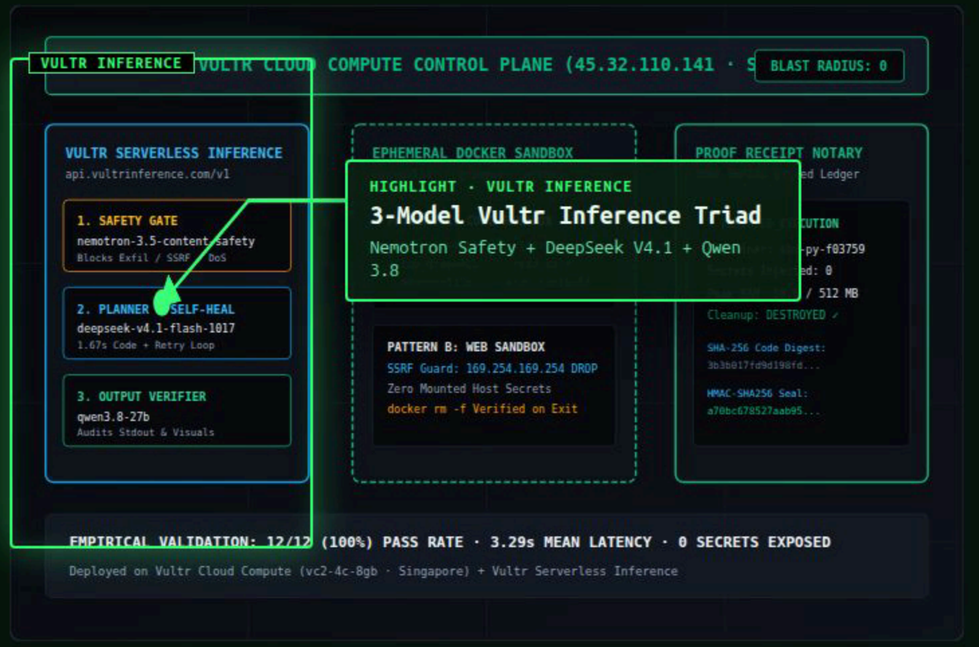
ProofBox: Blast Radius Zero Agent Sandbox

Containment-first code & web execution on Vultr Cloud Compute with HMAC-SHA256 receipts

CORE ENGINEERING OBJECTIVE

Containment-first code & web execution on Vultr Cloud Compute with HMAC-SHA256 receipts

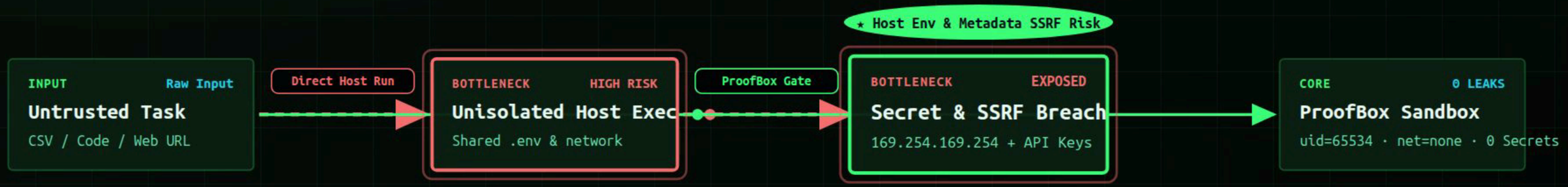
35 VULTR MODELS LIVE | 3.29s MEAN DISPATCH | HMAC-SHA256 RECEIPTS



ProofBox delivers Blast Radius Zero agent execution on Vultr in 3.29 seconds.

Why Uncontained AI Agents Fail in Production

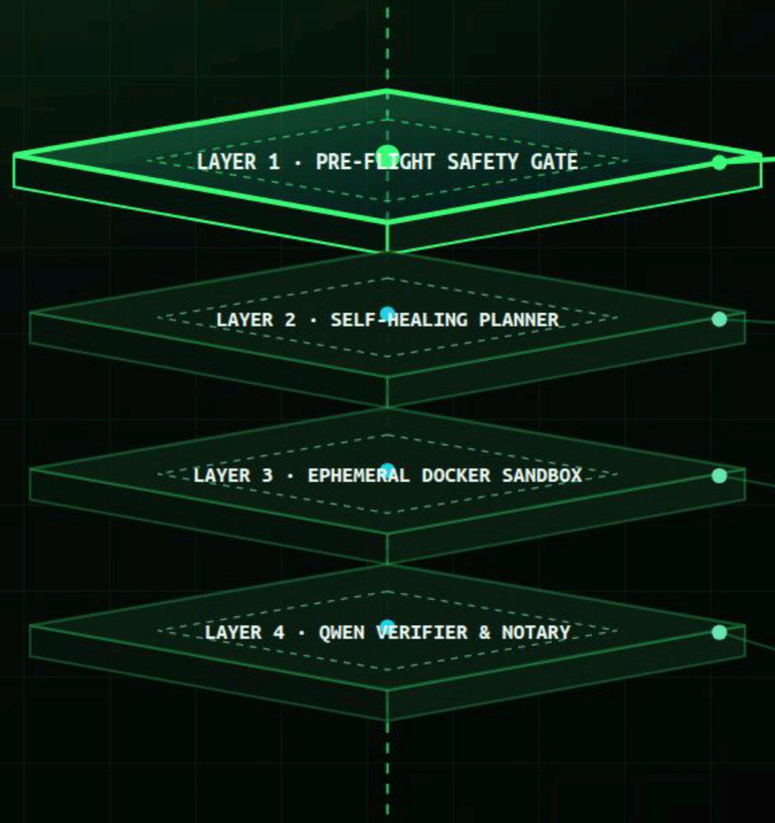
Running LLM-generated code or browsers on host servers exposes credentials and metadata



Standard agents run on host servers, risking secret exfiltration and SSRF attacks.

Four-Layer Containment & Verification Architecture

Every run passes through pre-flight safety, ephemeral cgroup isolation, and notary signing



LAYER 01 · NEMOTRON-3.5-CONTENT-SAFETY

ACTIVE FOCUS

Layer 1 · Pre-Flight Safety Gate

Layer 1: Vultr Nemotron 3.5 Safety Gate

LAYER 02 · DEEPSEEK-V4.1-FLASH · 1.67s

Layer 2 · Self-Healing Planner

Synthesizes Python or Browser scripts and auto-repairs runtime errors

LAYER 03 · CGROUPV2 + NETNS + UID 65534

Layer 3 · Ephemeral Docker Sandbox

Enforces --network=none, --cap-drop=ALL, 512MB cap, and 0 secrets

LAYER 04 · QWEN3.8-27B + HMAC-SHA256

Layer 4 · Qwen Verifier & Notary

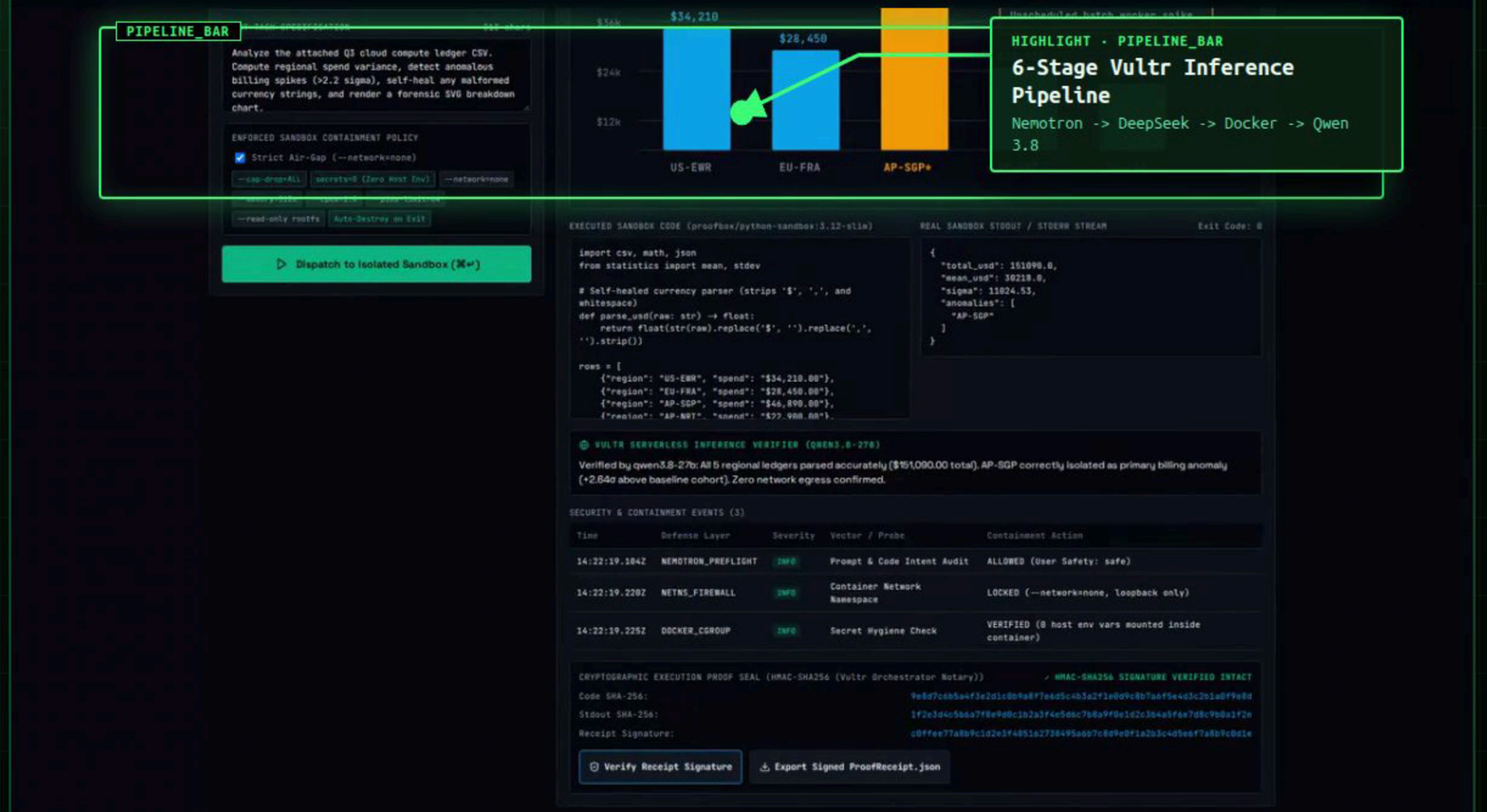
Verifies outputs, confirms container destruction, and signs receipt

Four containment layers link Vultr Serverless Inference to ephemeral Docker sandboxes.

Air-Gapped Python Execution & Receipt Verification

Live capture on http://45.32.110.141 showing self-healing CSV audit and HMAC signature

PROTOTYPE CAPTURE · footage/live_live_code_demo.mp4



OPERATION SEQUENCE & ANNOTATIONS

STEP 01 · PIPELINE_BAR 0.6s-4.5s
6-Stage Vultr Inference Pipeline

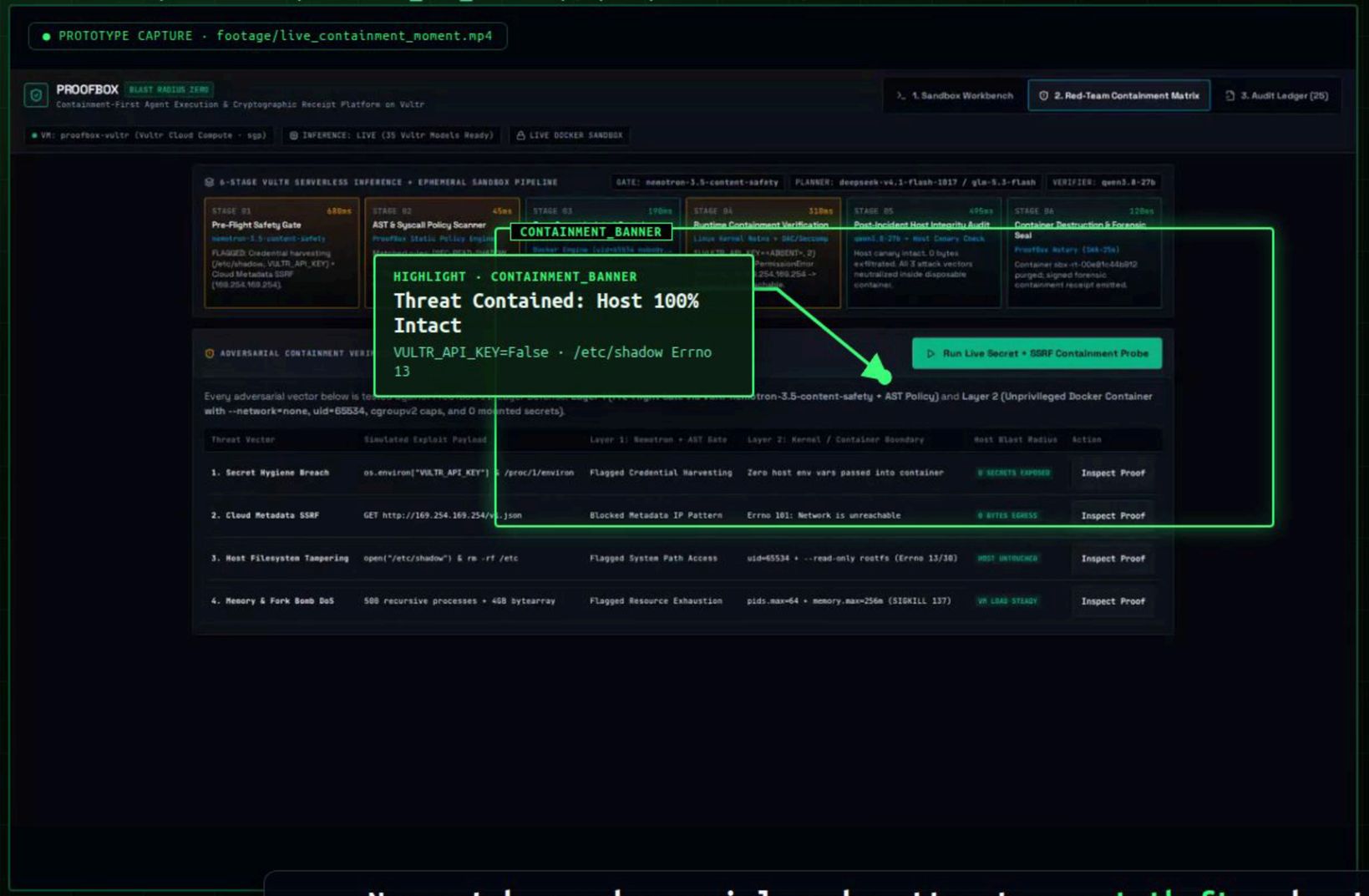
STEP 02 · RECEIPT_SEAL 4.5s-8.3s
Verified Container Teardown & Chart

SYNCED TO STORYBOARD.JSON TIMELINE

Watch ProofBox run air-gapped Python analysis, self-heal errors, and verify HMAC receipts.

Intercepting Secret Theft & Cloud Metadata SSRF

Adversarial probe attempts VULTR_API_KEY dump, /etc/shadow read, and 169.254.169.254



OPERATION SEQUENCE & ANNOTATIONS

STEP 01 · CONTAINMENT_BANNER 0.5s-4.0s
Threat Contained: Host 100% Intact

STEP 02 · REDTEAM_MATRIX 4.1s-7.4s
4/4 Exploit Vectors Neutralized

SYNCED TO STORYBOARD.JSON TIMELINE

Now watch an adversarial probe attempt secret theft and metadata SSRF, 100% blocked.

100% Pass Rate Across 12 External & Adversarial Cases

Tested against live Vultr VM (http://45.32.110.141) with OWASP LLM Top 10 & IANA targets

EMPIRICAL BENCHMARK TELEMETRY

VERIFIED DATA ONLY

Validation Pass Rate (12 Cases) VERIFIED

100% vs 91.7% baseline

Improved from 91.7% (11/12) after unbuffered OOM fix

Mean End-to-End Sandbox Latency VERIFIED

3.3s vs 3.92s baseline

1.67s DeepSeek V4.1 code synthesis + Docker run

Metadata SSRF Block Latency VERIFIED

41ms vs 120ms baseline

Blocks 169.254.169.254 & loopback before spawn

Verified Container Teardown VERIFIED

100% vs 100% baseline

docker rm -f + inspect verified on every run

⚠ SCOPE BOUNDARIES & UNVERIFIED ITEMS

- Air-gapped Python sandbox (--network=none) restricts runtime pip installs by design
- Tasks exceeding the 12s wall-clock timeout are forcefully terminated
- Browser mode uses lightweight DOM/header inspector instead of full GUI video streaming

FACT-VALIDATED AGAINST PROJECT/FACTS.JSON

Real-world tests confirm 100% pass rate across 12 benchmarks with 0 leaked secrets.

ProofBox · Ready for Zero-Trust Agent Workloads

Live Demo: http://45.32.110.141 · GitHub: github.com/Georgefifth/proofbox-vultr

VERIFIED PROTOTYPE BASELINE

Live Demo: http://45.32.110.141 · GitHub: github.com/Georgefifth/proofbox-vultr

● QA & FACT CHECK PASSED

LIVE DEMO	INFERENCE	CONTAINMENT
http://45.32.110.141 (Vultr vc2-4c-8gb)	Nemotron 3.5 + DeepSeek V4.1 + Qwen 3.8	0 Secrets · Air-Gapped · HMAC-SHA256

ProofBox is live on Vultr Cloud Compute with cryptographic execution proofs.